

Slide 1 – Title

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Smart Road Guardian+

Improving Pothole Detection Through Automatic Vehicle Monitoring

Presented by: _____

Slide 2 – The Problem

Potholes continue to be a serious challenge on South African roads. They damage vehicles, increase repair costs, contribute to road accidents, and make travelling unsafe. Although reporting platforms such as PotholeFixGP already exist, they depend largely on citizens manually reporting potholes. As a result, many potholes remain undetected or are reported long after they become dangerous.

Slide 3 – Our Solution

Our project, Smart Road Guardian+, improves the current pothole reporting process by introducing automatic detection. Instead of waiting for someone to report a pothole, a vehicle equipped with a vibration sensor detects the impact as it drives. A GPS module records the location, allowing the pothole to be identified quickly and accurately.

Slide 4 – How Our Prototype Works

Our prototype uses an Arduino Uno, an SW-420 vibration sensor, and a NEO-M8N GPS module. When the vibration sensor detects a strong impact, the GPS records the location of the vehicle. The system then creates a pothole report that is displayed on the Serial Monitor and is designed to be sent to a dashboard for municipal officials to review.

Slide 5 – The Dashboard

The dashboard displays the location of detected potholes together with the time they were recorded and their reporting status. This allows road maintenance teams to identify problem areas more quickly and plan repairs more efficiently. As more vehicles report the same location, municipalities gain greater confidence that the pothole requires attention.

Slide 6 – Benefits and Future Vision

Smart Road Guardian+ makes pothole detection faster, more reliable, and less dependent on manual reporting. By helping municipalities respond more quickly, the system has the potential to improve road safety, reduce vehicle damage, and make road maintenance more efficient. In the future, this technology could be installed in taxis, buses, municipal vehicles, and other fleet vehicles to monitor roads continuously across South Africa.

Slide 7 – Conclusion

Smart Road Guardian+ demonstrates how simple technology can improve existing road maintenance systems. Rather than replacing current reporting platforms, our solution enhances them by allowing vehicles to detect potholes automatically. Our vision is to create safer roads, reduce maintenance delays, and support smarter, more connected communities.

Thank you for listening. We welcome your questions.